Introduction to Artificial Neural Networks

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Motivation to Study Artificial Neural Networks (ANNs)

- Humans generally outperform computers in certain tasks (Haykin 1994):
  - Image and video recognition and processing {100-200 ms}
  - Voice recognition
  - Pattern recognition
  - Motor control
  - Heuristic optimization
- Traditional computers are good at tasks based on precise and fast arithmetic operations, outperforming biological systems (Churchland, 1986)

Source: Simon Haykin (2009)

What is Unique About Biological Systems?

“Method” of Information Processing

Biological Systems:

- Massively parallel
  - ~10 Billion “neurons” (nerve cells) and 60 trillion “synapses” for adults in the cerebral cortex (outer thin surface of brain)
- Analog in nature
- Adaptive and flexible
- 10^{-16} joules/operation/sec

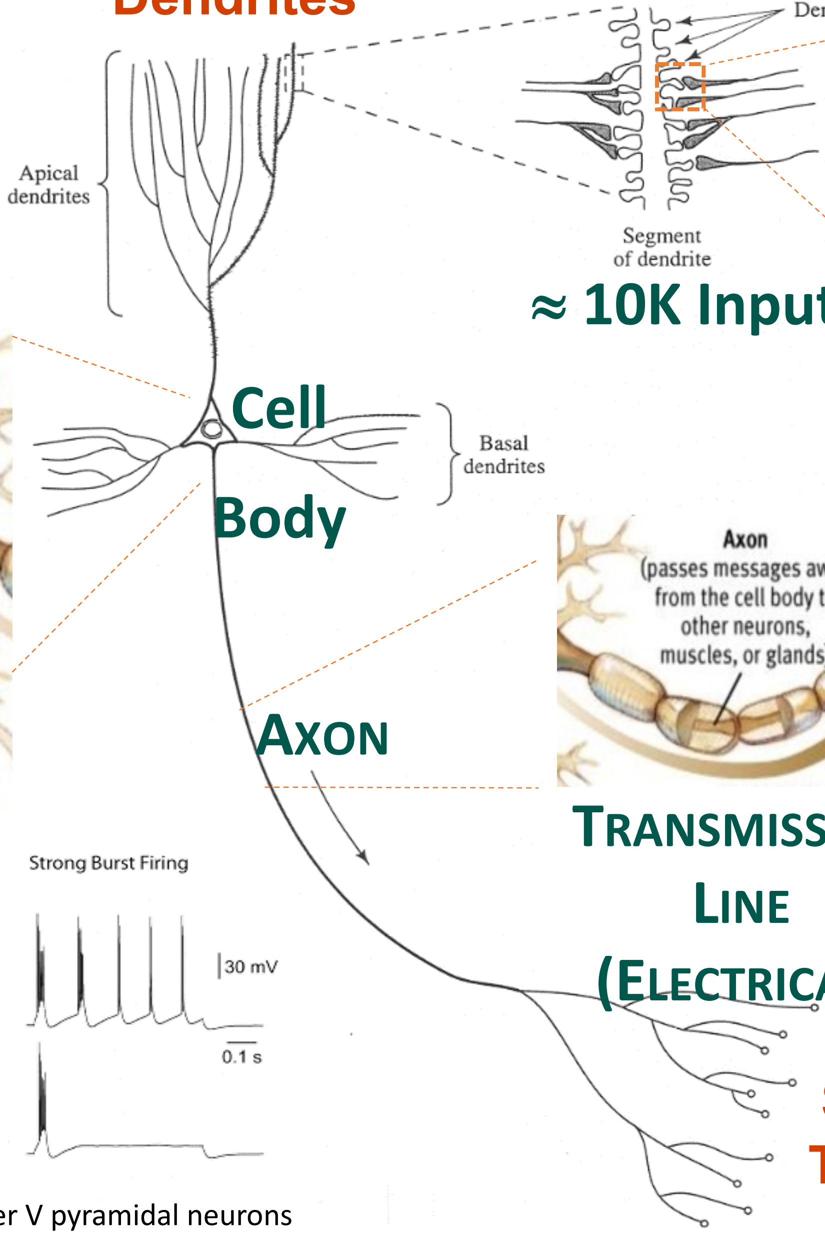
Computers:

- Serial machines
- Digital processing
- 10^{-6} joules/operation/sec (Foggin, 91)

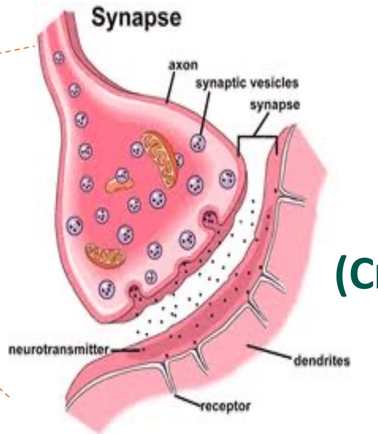
Source: Simon Haykin (2009)

“Pyramidal” Nerve Cell (Neuron)

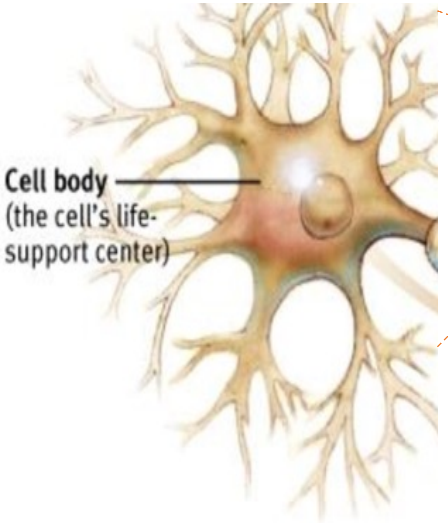
Dendrites



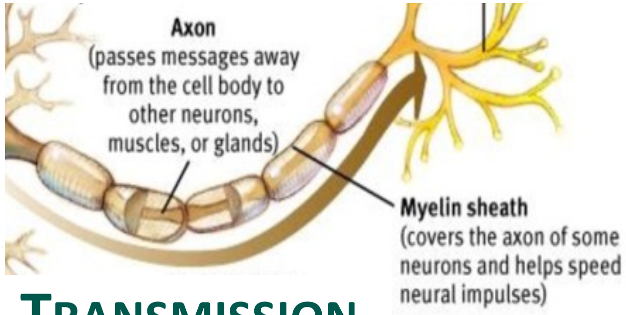
≈ 10K Inputs



(CHEMICAL)



Cell Body



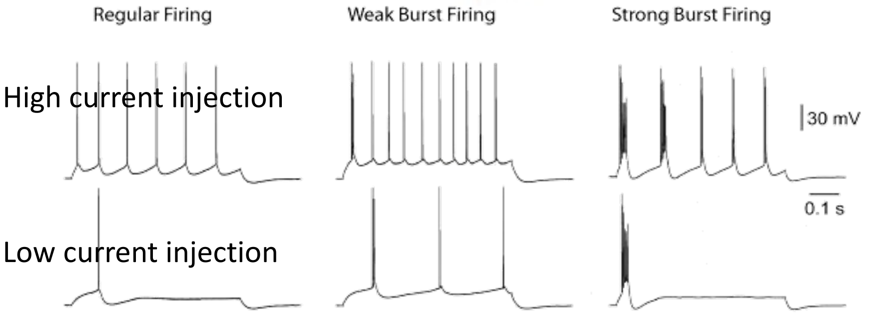
AXON

TRANSMISSION LINE (ELECTRICAL)

Synaptic Terminals

≈ 1K Outputs

Neurons produce “spike trains”



Cell responses of 3 different neocortical layer V pyramidal neurons
(Williams & Stuart, 1999)

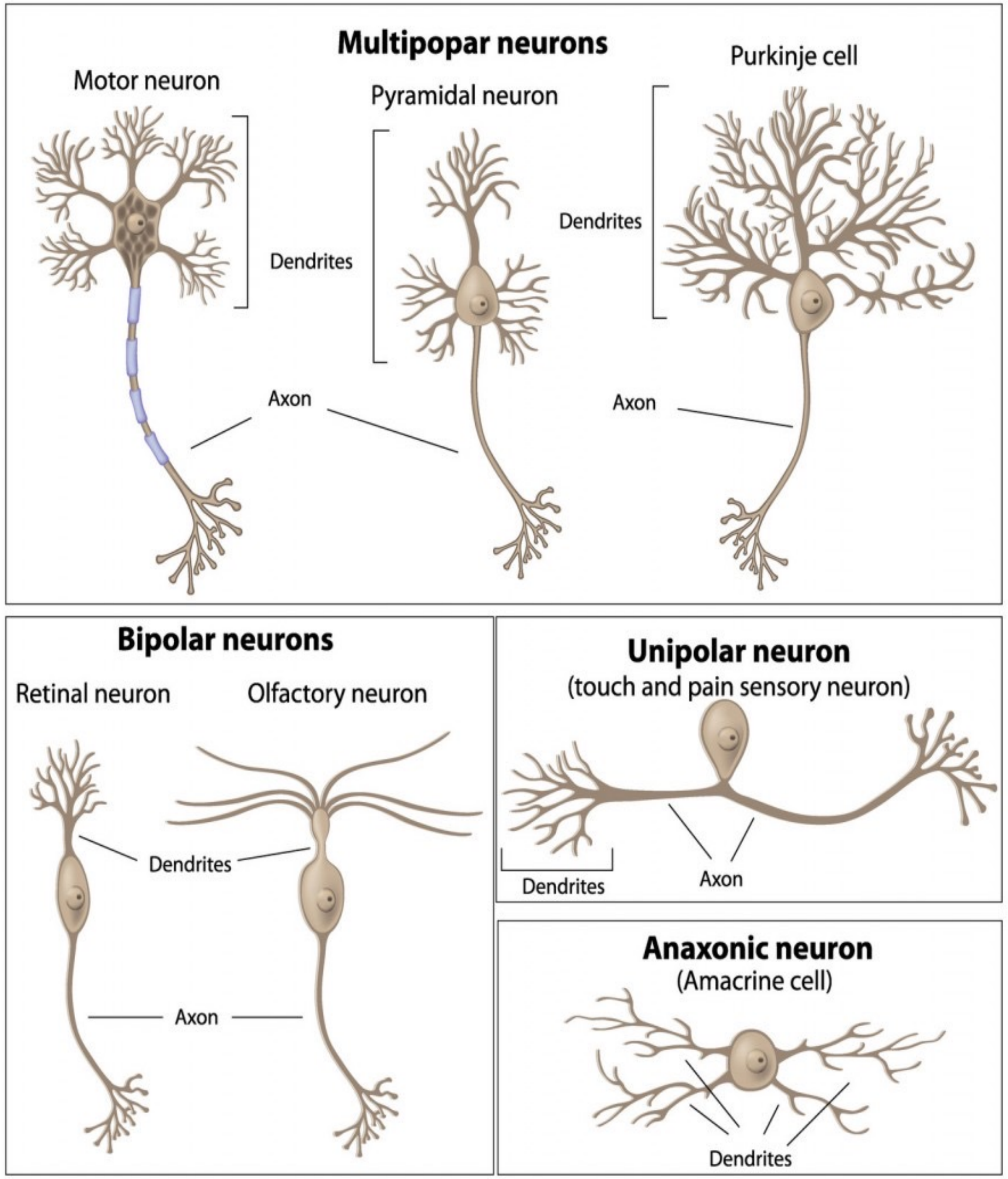
Different Types of Cells

CELLS

- Come in different sizes and shapes
- Exhibit plasticity

HUMAN CORTEX

- ~10 Billion Neurons
- ~60 Trillion Synapses



Basic Operation of a Real Neuron

- ***Neurons transmit signals via electrical pulses*** (action potentials or spike trains) ***along the axon***.
- ***These pulses reach downstream neurons at synapses***, primarily located on dendrites, which branch from the cell body (soma).
- Each pulse releases neurotransmitters that cross the synaptic cleft and bind to post-synaptic receptors on the dendrites.
- The binding changes dendritic membrane potential, ***producing a post-synaptic potential*** (PSP).
- ***PSPs can either inhibit (Hyperpolarize) or excite (Depolarize) downstream neuron***.
  - Size and type of PSP depend on factors like synapse geometry and neurotransmitter type.
- ***PSPs travel along dendrites, spreading over the soma to the axon hillock.***
- ***Neuron integrates thousands of PSPs over time, and if the potential at the axon hillock exceeds a threshold, the neuron fires an action potential.***
- ***This action potential then travels along the axon, initiating the process in other neurons in the efferent pathway.***

(Source: Dr. Kevin Gurney, Psychology Department, University of Sheffield)

Cerebral Cortex

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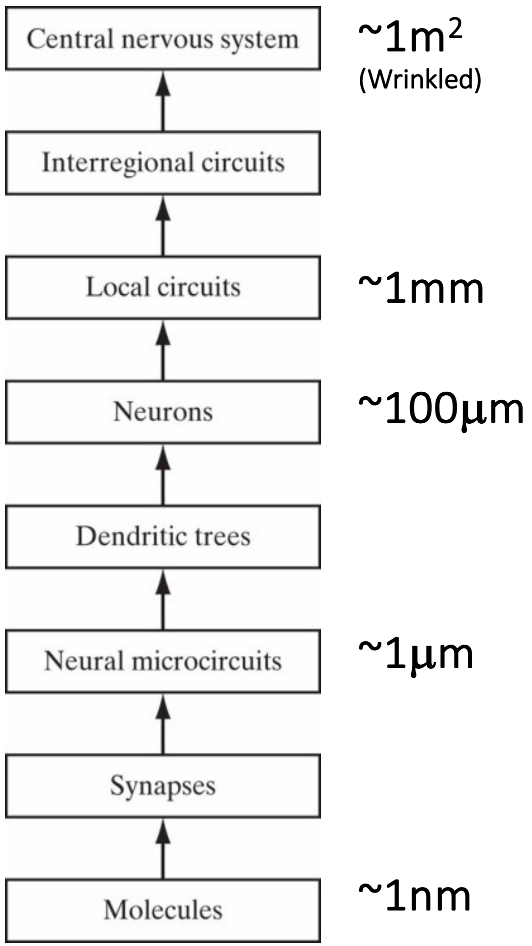
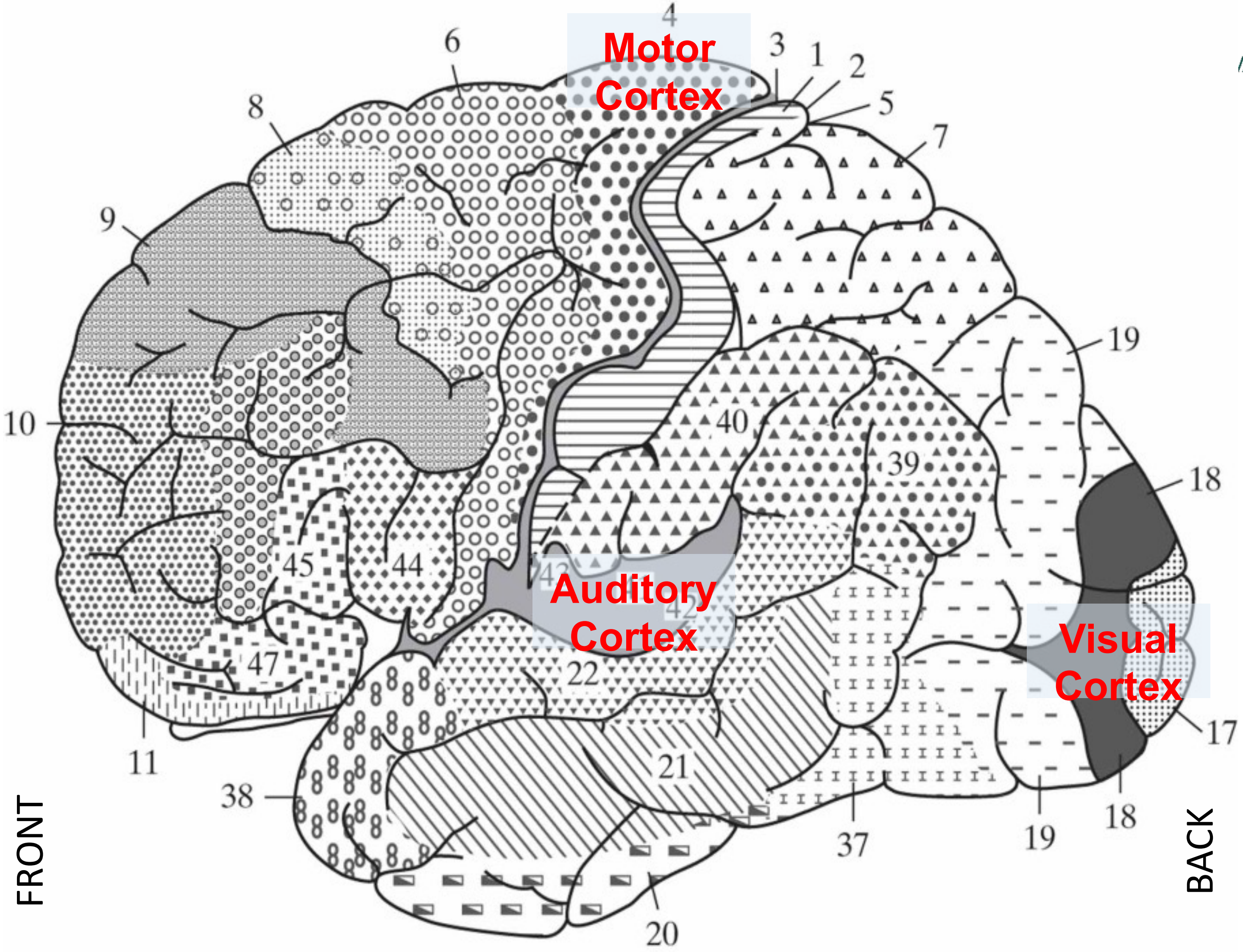


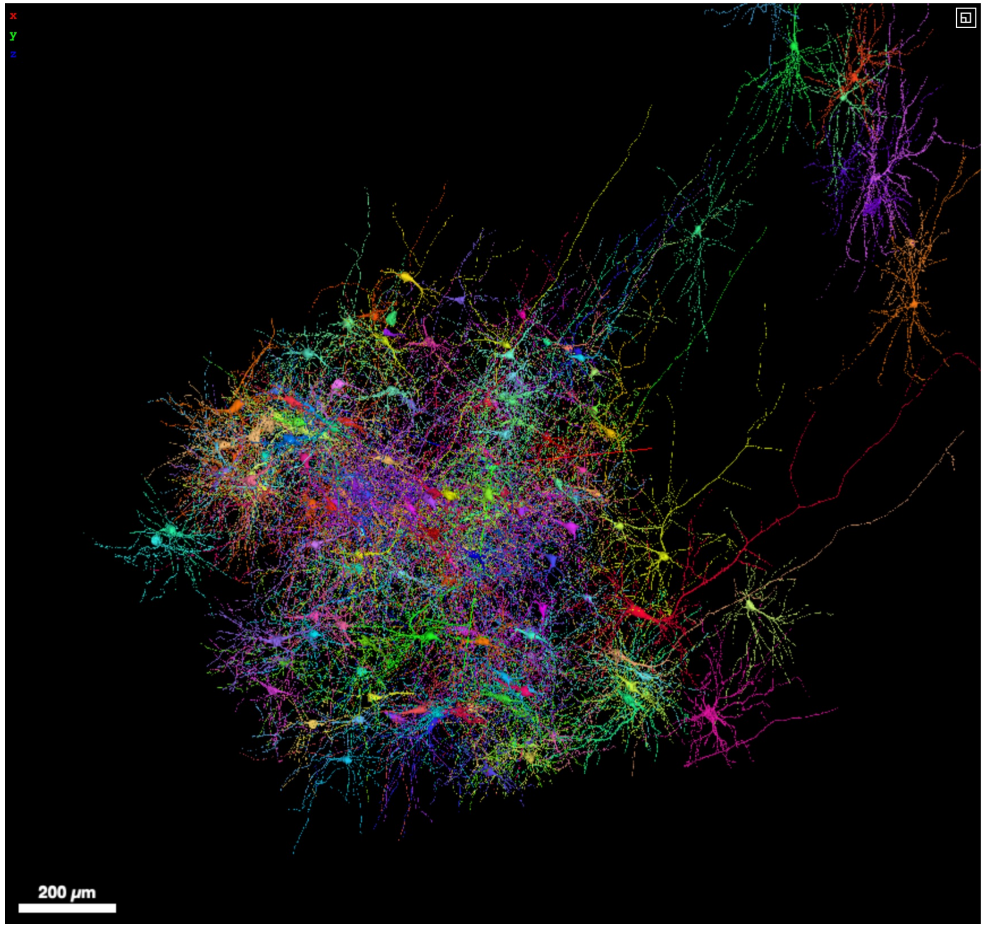
Structural Organization of Levels in the Brain

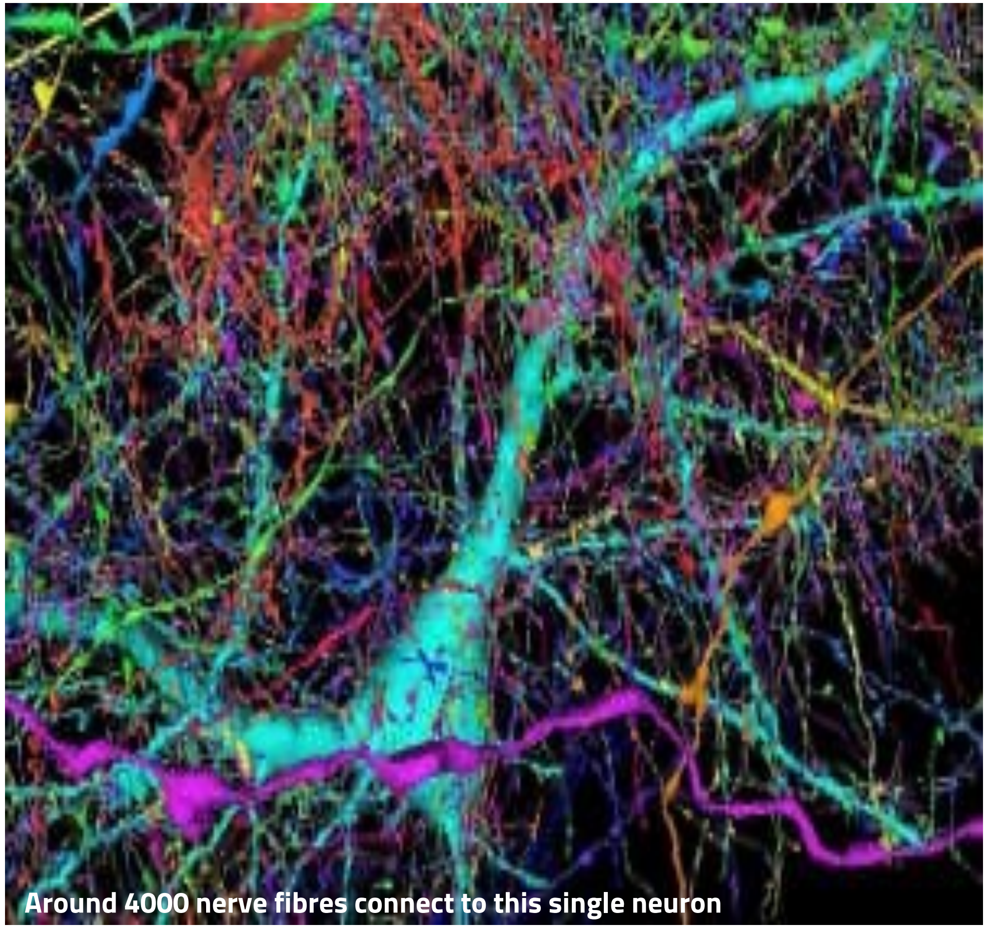


Different areas identified by thickness of layers and types of cells within. Some key sensory areas are as follows: Motor cortex: motor strip, area 4; premotor area, area 6; frontal eye fields, area 8. Somatosensory cortex: areas 3, 1, and 2. Visual cortex: areas 17, 18, and 19. Auditory cortex: areas 41 and 42.

Mapping of a Piece of Human Brain (Google)

- Partial brain map, available online, includes 50k cells
  - Joined together by 130M synapses. Dataset measures 1.4 petabytes.





Source: Marshall (2021) | Link

Results from Efforts to Imitate Biology (last century)

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- Advances in applying ANNs for problems found intractable or difficult for traditional computation
  - Examples: Pattern recognition (e.g., sensors, images, video, audio), classifiers, function approximation (e.g., regression), and forecasting.
- ANNs can supplement processing by Von Neuman digital computer
- Tip of the "ice-berg"
  - Most ANN’s are being simulated on serial machines
  - Most ANNs have 1~2k nodes vs 10 billion neurons and 60 trillion synapses
  - Simplistic structures/connectivity
  - Simplistic neuron models
  - Simplistic learning algorithms

Recent “Transformer” Models are Monstrous

- GPT3 from OpenAI: Trained on 570GB of Data | Link
- Wu Dao 2.0 (2021): 1.75T Parameters, Trained on 4.9TB of High-Quality Text & Images | Link

(Founding Father: Raymon Cajal, 1911)

Characteristics of Most ANNs

- Function as **parallel** distributed computing networks
- Most basic characteristic is **architecture**
- Provide either “**instantaneous**” or “**dynamic**” responses
- **Must be taught or trained** (supervised or unsupervised learning)
- Resort to different “**learning algorithms**” and “**learning modes**”
- **Learn patterns**, functional dependencies
- Employ several types of **neuron models**

Source: Simon Haykin (2009)

Benefits of ANNs

- Non-linearity
- Non-parametric
  - Nonparametric in that they make minimal assumptions about the underlying "distribution" of the data
  - ANNs do have parameters (e.g., synaptic weights)
- Input-Output Mapping or Approximation
- VLSI (Silicon Hardware) Implementability
- Graceful degradation if implemented in hardware
- Uniformity of Analysis and Design
- Adaptivity while maintaining overall system "stability"
  - Stability-Plasticity dilemma (Grossberg, 1988)

Weakness: General lack of confidence intervals

Source: Simon Haykin (2009)

Machine Surpasses Human-Level Image Classification

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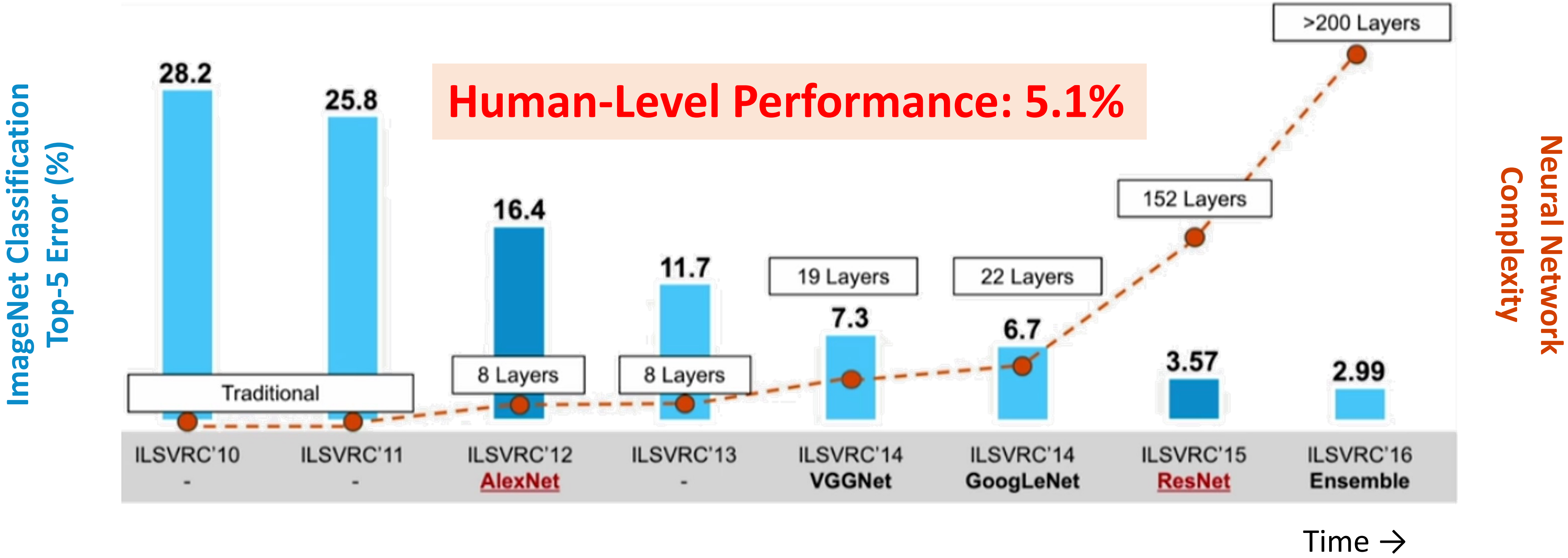
- IMAGENET:
  - Image Classification Challenge
  - 1,000 Object "Classes"
  - 1,431,167 Images



Top-5 error is the fraction of test samples where the correct label does not appear in the top 5 predicted results of the model when results are sorted in decreasing order of confidence.

IMAGENET Source: Link

Deep Learning for Image Processing

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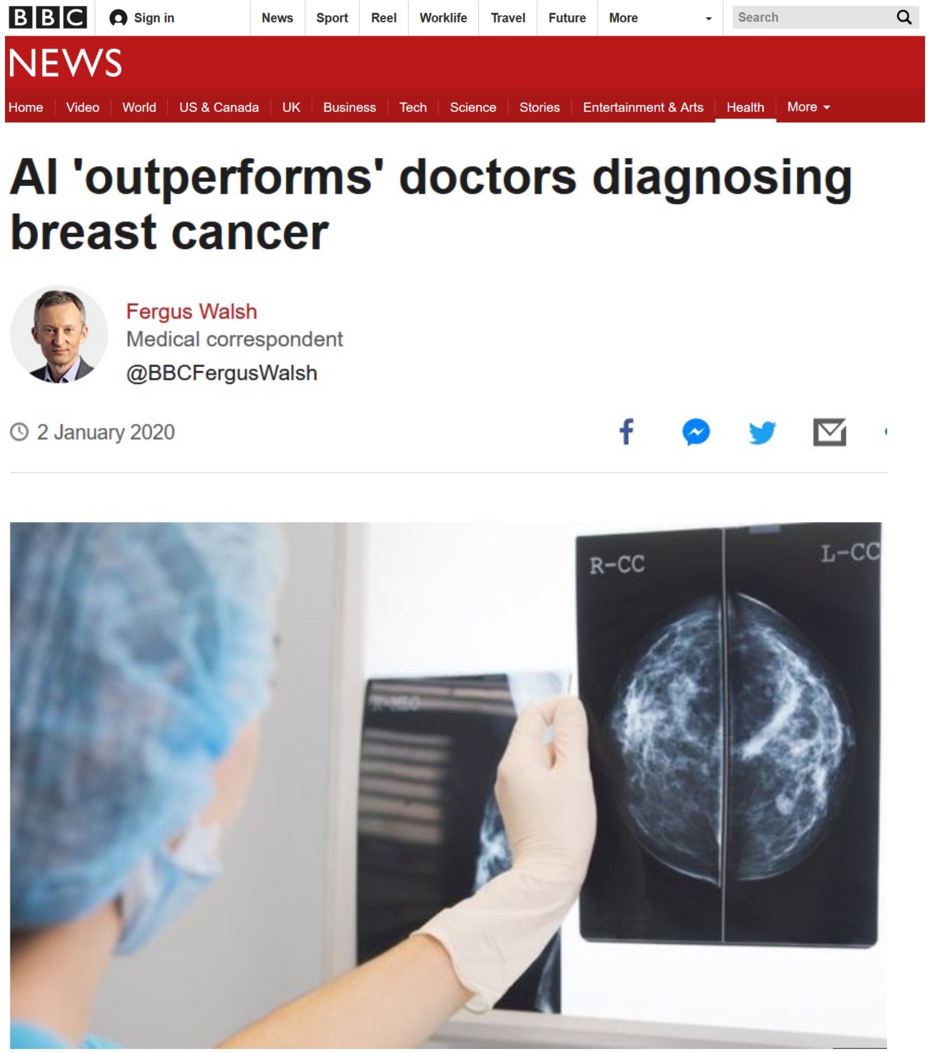
- **AI is more accurate than doctors in diagnosing breast cancer from mammograms, a study in the journal Nature suggests.**
- An international team, including researchers from Google Health and Imperial College London, designed and trained a computer model on X-ray images from nearly 29,000 women.
- Algorithm outperformed six radiologists in reading mammograms.
- AI was still as good as two doctors working together.
- Unlike humans, AI is tireless. Experts say it could improve detection.



Source: BBC Article | Jan 2020 Nature Article

FDA Approves DL for Breast Cancer Detection

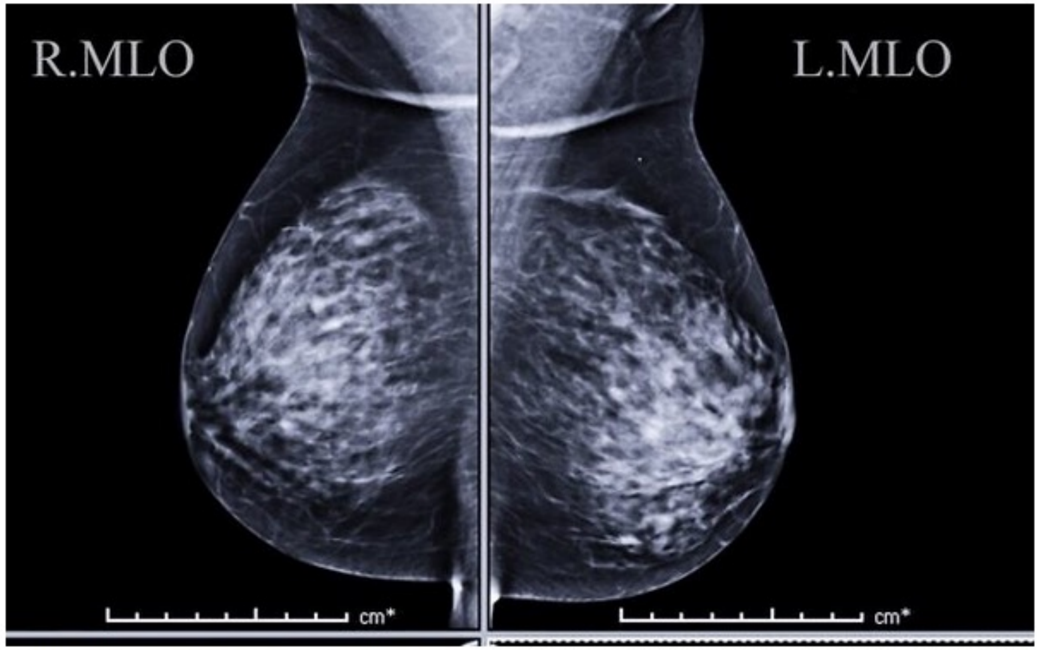
- Zebra Medical Vision, an Israel kibbutz-based startup that uses artificial intelligence technology to help read medical scans, said it has received US Food and Drug Administration (FDA) clearance for use of its algorithm to read mammograms.
- The FDA nod follows the European CE approval for its “HealthMammo” mammography solution, which supports radiologists in their search for breast tumors by prioritizing and identifying suspicious mammograms.

Israel’s Zebra gets FDA nod for AI-based software that helps spot breast cancer

‘HealthMammo’ solution aims to support radiologists in their tumor diagnoses by identifying and prioritizing suspicious mammograms

By SHOSHANNA SOLOMON
27 July 2020, 3:00 pm | 0

253 shares



An illustrative image of a mammogram (Courtesy)

Source: The Times of Israel | [Article]

Managing Medical Uncertainty

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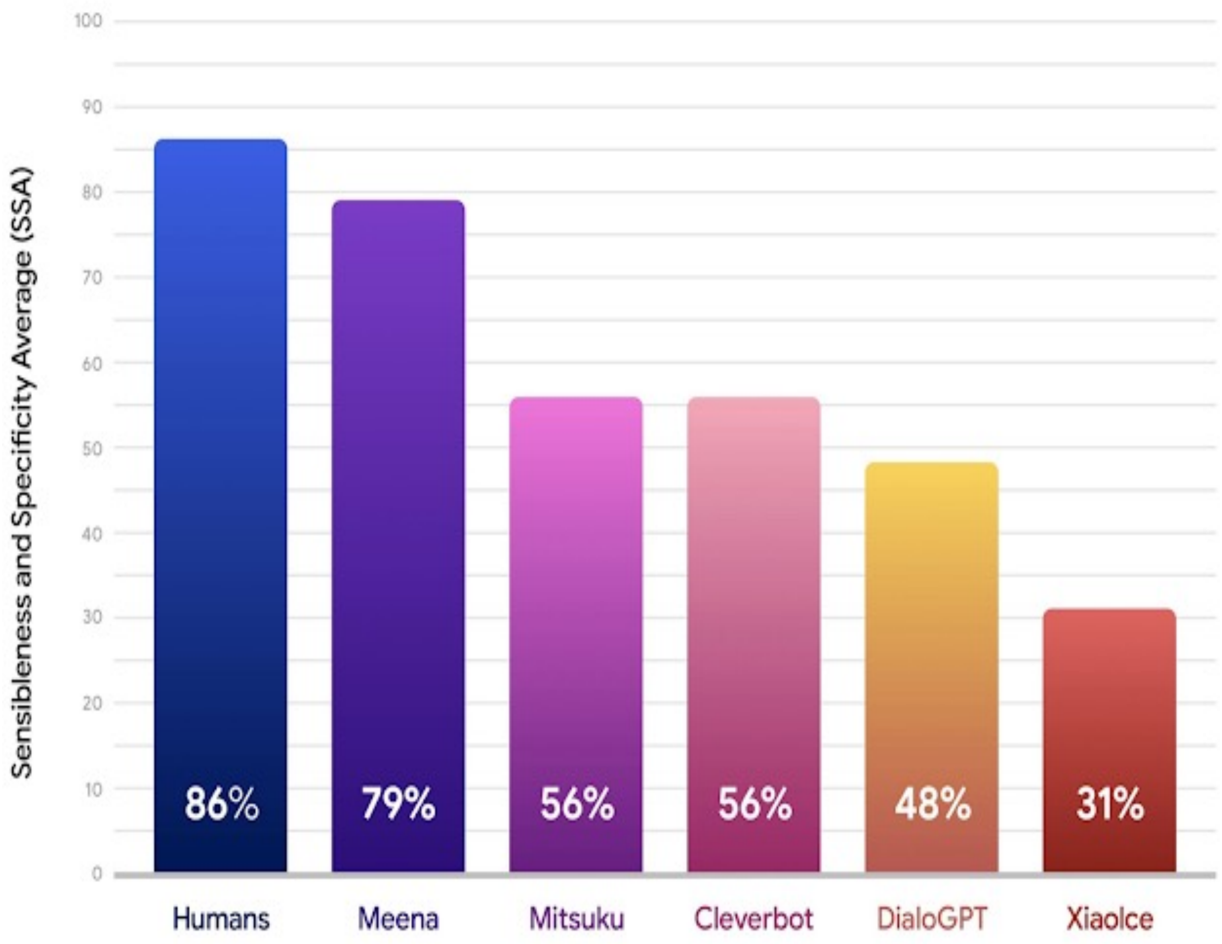
- Doctors are using ML systems to ***assess patient risk***, *The Wall Street Journal* reported.
- Duke uses ***Sepsis Watch*** to monitor ER patients for acute inflammation, which causes one in three hospital deaths. It analyzes 86 variables every 5 minutes and alerts nurses if necessary.
- Kaiser Permanente deployed Advanced Alert Monitor in 21 hospitals after finding it reduced hospital stays and ICU referrals. ***It predicts ICU needs*** within 12 hours based on vital signs, test results, and coexisting conditions.
- University of Maryland Medical System found that a ML model outperformed traditional methods in ***predicting a patient's 30-day readmission risk***.
- In April 2022, the EU approved an AI system that ***scans chest x-rays and writes reports*** for patients with no discernable issues.
- In October 2021, Canada, the UK, and the US jointly issued ***guiding principles for ML in medicine***.
- In November 2020, U.S. Medicare and Medicaid agreed to reimburse doctors using two AI-powered tools: Viz LVO, which ***monitors patients for signs of a stroke***, and IDx-DR, which ***helps diagnose a complication of diabetes that can cause blindness***.

Source: The Batch (May 2022) | Newsletter

AI Chatbots: Towards a Conversational Agent that Can Chat About…Anything

- In "Towards a Human-like Open-Domain Chatbot", we present Meena, a 2.6 billion parameter end-to-end trained neural conversational model.
- For each chatbot, we collect between 1600 and 2400 individual conversation turns through about 100 conversations. Each model response is labeled by crowdworkers to indicate if it is sensible and specific.
- **We show that Meena can conduct conversations that are more sensible and specific than existing state-of-the-art chatbots.**



Source: Google AI Blog | Jan 2020 Blog

ChatGPT from OpenAI.com

chat.openai.com

NYT CNBC BBC Fid CHASE PPal Citi Agile WSU Mail Gmail WSU ISE Sch GS A C IR Cuse DSBA Big Data

+ New chat

Detection Of Data Ano

Clear conversations

Dark mode

OpenAI Discord

Updates & FAQ

Log out

What are the different types of data anomalies

There are several types of data anomalies, also known as outliers, that can occur in a dataset:

1. Point anomalies: These are individual data points that deviate significantly from the rest of the data. For example, a single data point in a time series that is significantly higher or lower than the surrounding data points.
2. Contextual anomalies: These are data points that are anomalous in a specific context or within a specific subset of the data.
3. Collective anomalies: These are groups of related data points that deviate from the rest of the data as a whole.
4. Event-based anomalies: These are abnormal patterns or events that occur at a specific point in time.
5. Collective event-based anomalies: These are groups of related abnormal patterns or events that occur at a specific point in time.
6. Spatial anomalies: These are data points that deviate from the rest of the data in a spatial context.
7. Collective spatial anomalies: These are groups of related data points that deviate from the rest of the data in a spatial context.
8. Sequential anomaly: These are anomalies that occur in sequence or over time.

It is important to note that the distinction between these different types of anomalies may vary depending on the domain and the data set under consideration.

Source: ChatGPT from OpenAI | Jan 21, 2023 | [Link]

Give Python code for change point detection

Change point detection is the process of identifying the point in time at which the statistical properties of a time series change. Here's an example of how to perform change point detection in Python using the `ruptures` library:

Copy code

```
!pip install ruptures
import ruptures as rpt

# generate data
n_samples, dim, sigma = 1000, 3, 4
n_bkps = 4  # number of breakpoints
signal, bkps = rpt.pw_constant(n_samples, dim, n_bkps,
noise_std=sigma)

# detection
algo = rpt.Pelt(model='rbf').fit(signal)
result = algo.predict(pen=10)

print(result)
```

The above code uses the `Pelt` algorithm from the `ruptures` library to detect change points in a synthetic data set. The `fit` method is used to fit the model to the data, and the `predict` method is used to find the change points. The `result` variable contains the indices of the detected change points in the data.

There are several other algorithms that can be used for change point detection, such as `Binseg`, `Dynp`, and `Window`. Each algorithm has its own strengths and weaknesses, and the best one to use depends on the specific characteristics of the data.

You can also use other libraries like `changepy` or `prophet` for change point detection on time series data.

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LLMs: Anthropic Ups the Ante

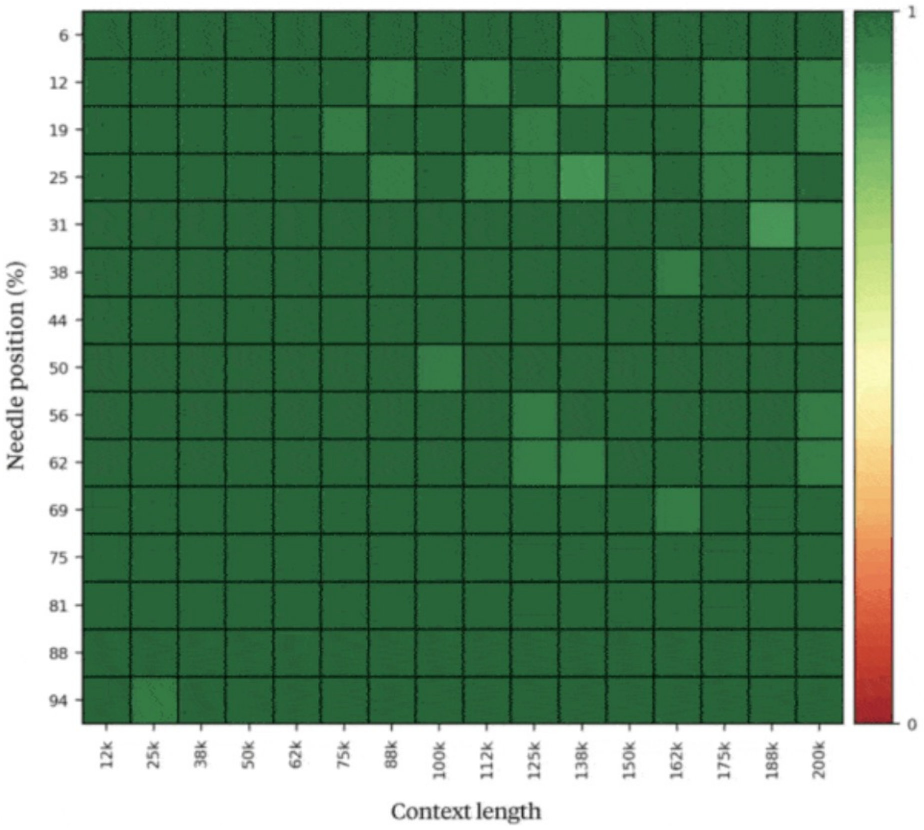
	Claude 3 Opus	Claude 3 Sonnet	Claude 3 Haiku	GPT-4	GPT-3.5	Gemini 1.0 Ultra	Gemini 1.0 Pro
Undergraduate level knowledge *MMLU*	86.8% 5 shot	79.0% 5-shot	75.2% 5-shot	86.4% 5-shot	70.0% 5-shot	83.7% 5-shot	71.8% 5-shot
Graduate level reasoning *GPQA, Diamond*	50.4% 0-shot CoT	40.4% 0-shot CoT	33.3% 0-shot CoT	35.7% 0-shot CoT	28.1% 0-shot CoT	—	—
Grade school math *GSM8K*	95.0% 0-shot CoT	92.3% 0-shot CoT	88.9% 0-shot CoT	92.0% 5-shot CoT	57.1% 5-shot	94.4% Maj1@32	86.5% Maj1@32
Math problem-solving *MATH*	60.1% 0-shot CoT	43.1% 0-shot CoT	38.9% 0-shot CoT	52.9% 4-shot	34.1% 4-shot	53.2% 4-shot	32.6% 4-shot
Multilingual math *MGSM*	90.7% 0-shot	83.5% 0-shot	75.1% 0-shot	74.5% 8-shot	—	79.0% 8-shot	63.5% 8-shot

	Claude 3 Opus	Claude 3 Sonnet	Claude 3 Haiku	GPT-4	GPT-3.5	Gemini 1.0 Ultra	Gemini 1.0 Pro
Code *HumanEval*	84.9% 0-shot	73.0% 0-shot	75.9% 0-shot	67.0% 0-shot	48.1% 0-shot	74.4% 0-shot	67.7% 0-shot
Reasoning over text *DROP, F1 score*	83.1 3-shot	78.9 3-shot	78.4 3-shot	80.9 3-shot	64.1 3-shot	82.4 Variable shots	74.1 Variable shots
Mixed evaluations *BIG-Bench-Hard*	86.8% 3-shot CoT	82.9% 3-shot CoT	73.7% 3-shot CoT	83.1% 3-shot CoT	66.6% 3-shot CoT	83.6% 3-shot CoT	75.0% 3-shot CoT
Knowledge Q&A *ARC-Challenge*	96.4% 25-shot	93.2% 25-shot	89.2% 25-shot	96.3% 25-shot	85.2% 25-shot	—	—
Common Knowledge *HellaSwag*	95.4% 10-shot	89.0% 10-shot	85.9% 10-shot	95.3% 10-shot	85.5% 10-shot	87.8% 10-shot	84.7% 10-shot

Claude 3 Opus

Recall accuracy over 200K

(averaged over many diverse document sources and 'needle' sentences)



Source: The Batch | March 13, 2024 | LINK

Is AI Better than Doctors in Answering Questions?

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- **Study**: Researchers collected 195 questions and answers from anonymous users of a social media site posed to doctors who volunteer to answer.
- Questions were later submitted to ChatGPT.
- A panel of three physicians or nurses then rated both sets of answers for quality and empathy.
- **What did the study find?**
  - For nearly 80% of answers, ChatGPT was considered better than the physicians.
  - **Good or very good quality answers**: ChatGPT received these ratings for 78% of responses, while physicians only did so on 22% of responses.
  - **Empathetic or very empathetic answers**: ChatGPT scored 45% and physicians 4.6%.

- **Study Limitations**
  - **Evaluating and comparing answers**: Evaluators applied untested, subjective criteria for quality and empathy.
  - **Difference in length of answers**: More detailed answers from ChatGPT might seem to reflect patience or concern.
  - **Incomplete blinding**: AI-generated communication does not always sound exactly like a human, and AI answers were longer. So, it's likely that for at least some answers, the evaluators were not blinded.

STAYING HEALTHY

Can AI answer medical questions better than your doctor?

A second look at a study rating quality and empathy when answering patient questions.

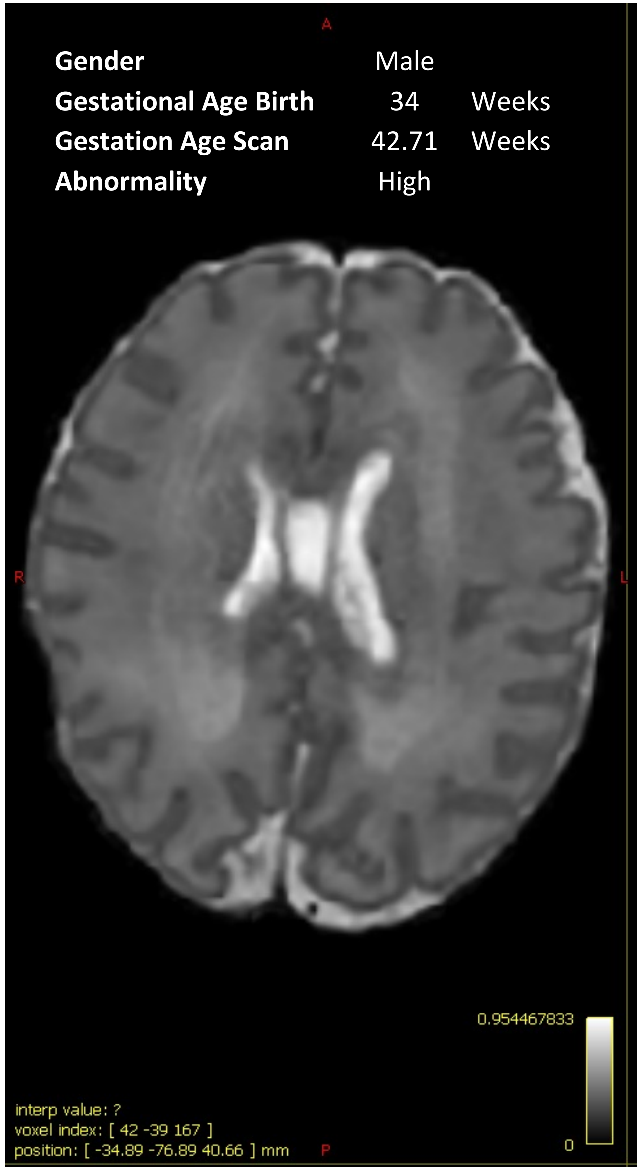
March 27, 2024

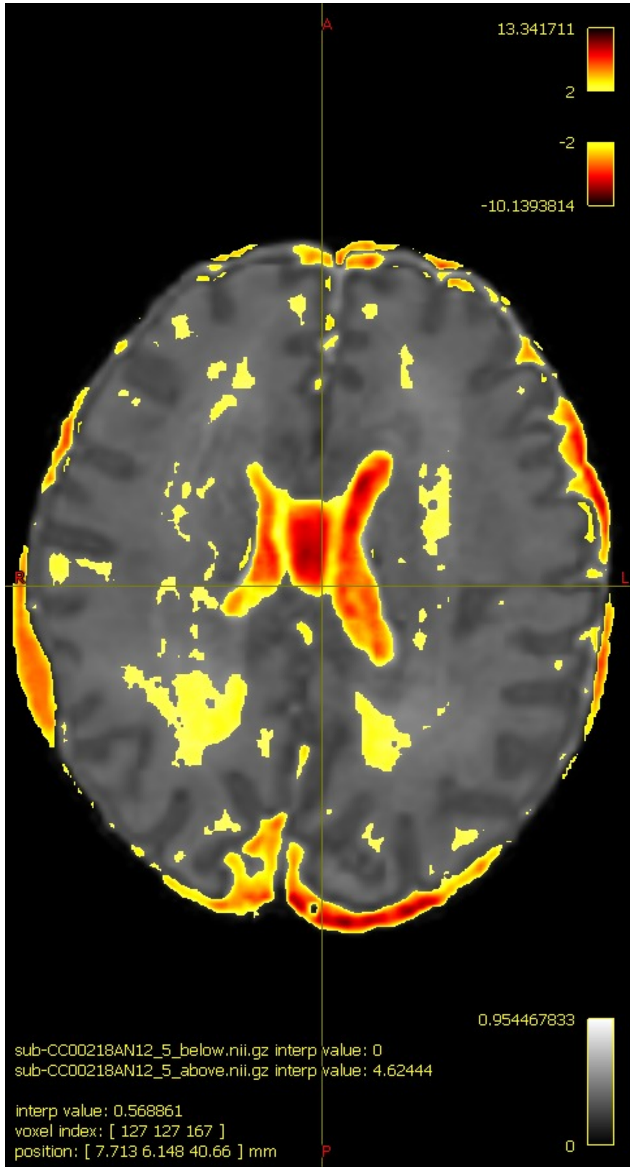
By Robert H. Shmerling, MD, *Senior Faculty Editor, Harvard Health Publishing; Editorial Advisory Board Member, Harvard Health Publishing*

Source: Dr. Shmerling (March 2024) | Article

Variational Auto-Encoders for MRI Anomaly Detection

- **Setting**: Neonatal MRI Images
- **Data Source**: Developing Human Connectome Project (dHCP)
  - King's College London, Imperial College London and U. of Oxford
- **Collaboration**: Wayne State University & Children's Hospital of Michigan
  - WSU: Jad Raad & Drs. Chinnam, Arslanturk
  - CHM: Drs. Tan, Mody, Jeong
- **Focus**: Automated Anomaly Detection
- **Approach**: Deep Learning Variational Auto-Encoders
- **Preliminary Results are Very Promising!**





Deep Learning for Precision Medicine

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‘IT WILL CHANGE EVERYTHING’: DeepMind’s AI makes gigantic leap in solving protein structures

- AlphaFold: Google’s deep-learning program for determining the 3D shapes of proteins stands to transform biology, say scientists.
- “I think it’s fair to say this will be very disruptive to the protein-structure-prediction field. It’s a breakthrough of the first order, certainly one of the most significant scientific results of my lifetime.”
  - Mohammed AlQuraishi, Computational Biologist, Columbia University
- **“It gave us the structure for a bacterial protein in half an hour, after we had spent a decade trying everything,”** Lupas says.
  - Andrei Lupas, Evolutionary Biologist, Max Planck Institute for Developmental Biology

A protein’s function is determined by its 3D shape.

Source: Nature | Nov 2020 Article

AI for Art & Marketing

- The latest experiment in [AI art direction] comes from Nestlé yogurt and dessert brand La Laitière.
- The campaign uses Dall-E's new Outpainting feature, which can generate visuals that expand the borders of a picture.
- When it launched earlier this month, Dall-E demonstrated Outpainting's capabilities with Johannes Vermeer's famous painting, "Girl with a Pearl Earring"—expanding the background to show the subject in her imagined home.
- La Laitière put Outpainting to the test with another Vermeer painting, "The Milkmaid"—fitting, since the brand shares a name with the artwork in French.
- Created by Ogilvy Paris, the ad extends the world of The Milkmaid using the AI art direction tool, generating new imagery that tells a broader story that is in keeping with the Dutch artist's style. Beyond the original frame of Vermeer's oil-on-canvas painting, the kitchen maid is shown working in a bustling household, while children play underfoot, and noblemen and women wait eagerly for her to prepare some milky La Laitière-inspired treats.
- "With this 'extended' Laitière, we are at the very beginning of a revolution for our creative industry. AIs represent an incredible potential that impacts the way we conceive, design, produce and do justice to realize the idea's fullest potential," David Raichman, executive creative director, social and digital at Ogilvy Paris, said in a statement.

Nestlé Brand Is Latest to Venture Into Brave New World of AI Art Direction

The campaign is part of a trend using image generators to create ads

By Brittaney Kiefer | Sep 21, 2022 | 9:33 AM

The debate over whether robots could one day take over creative jobs has been raging for years now, but recent advances in artificial intelligence (AI) art has opened up a new playground for advertisers. Earlier this year, Heinz made waves when it used research group OpenAI's machine learning algorithm, Dall-E 2, in a marketing stunt. The food company fed generic ketchup-related prompts into the art generator and it conjured up images of Heinz bottles, demonstrating the ubiquity of the brand.

Source: Apple News| Sep 2022 | Link

Deep Learning in Banking

- **JPMorgan Chase rolled out a generative AI assistant** to tens of thousands of its employees, the initial phase of a broader plan to inject the technology throughout the bank.
- The program, called LLM Suite, is already **helping more than 60,000 employees with tasks like writing emails and reports**.
- The software is expected to eventually be as ubiquitous within the bank as the videoconferencing program Zoom.
- JPMorgan designed **LLM Suite to be a portal that allows users to tap external large language models** and launched it with ChatGPT maker OpenAI's LLM.

JPMorgan Chase is giving its employees an AI assistant powered by ChatGPT maker OpenAI

HUGH SON | PUBLISHED FRIDAY, AUG 9, 2024

JPMorgan Chase has rolled out a generative artificial intelligence assistant to tens of thousands of its employees in recent weeks, the initial phase of a broader plan to inject the technology throughout the sprawling financial giant.

The program, called LLM Suite, is already available to more than 60,000 employees, helping them with tasks like writing emails and reports. The software is expected to eventually be as ubiquitous within the bank as the videoconferencing program Zoom, people with knowledge of the plans told CNBC.

Rather than developing its own AI models, JPMorgan designed LLM Suite to be a portal that allows users to tap external large language models — the complex programs underpinning generative AI tools — and launched it with ChatGPT maker OpenAI's LLM, said the people.

"Ultimately, we'd like to be able to move pretty fluidly across models depending on the use cases," Teresa Heitsenrether, JPMorgan's chief data and analytics officer, said in an interview. "The plan is not to be beholden to any one model provider."

Source: CNBC | AUG 2024 Article

Financial firm already saves $25 million by using AI

- You can count billionaire investor Steve Cohen among those who believes AI is already making an impact on businesses.
- The Point72 founder told CNBC’s Andrew Ross Sorkin on “Squawk Box” that his financial firm has found ways for even the early AI models to save the company money.
- “I’ll give you one little anecdote. My CTO comes to me and says I can save the firm $25 million by using these LLMs to improve our efficiency,” Cohen said, referencing the large language models like ChatGPT.
- “Now, we’re a nice sized firm. We’re not a huge firm. So, imagine what big companies can do. And that’s just one thing, so it gives you a little bit of a look into what’s possible”.
- Cohen called AI a “really durable theme” for investing and said that basically every company needs to be thinking about how it can change the business.
- “If you’re a company and not thinking about this, you’re going to wake up one day and go ‘we’re in trouble,’” Cohen said.

MARKETS

Steve Cohen says his financial firm can already save $25 million by using AI

PUBLISHED WED, APR 3 2024•12:15 PM EDT

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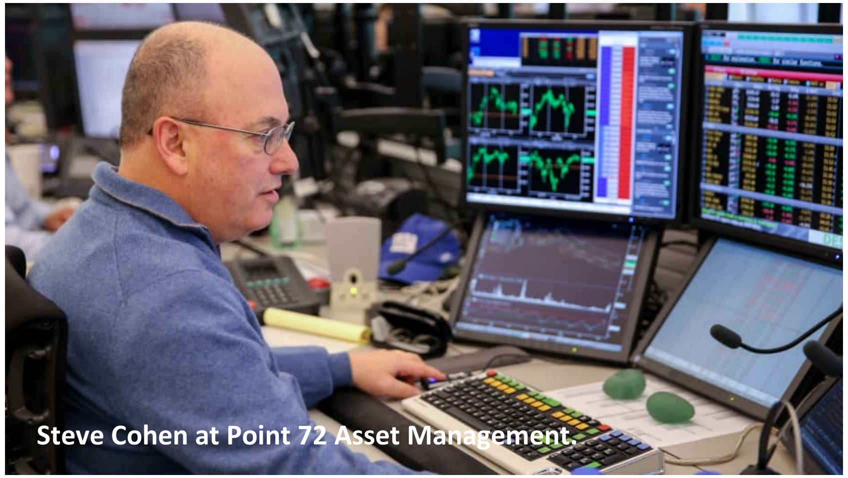


Steve Cohen at Point 72 Asset Management.

BMW uses A.I. to Improve Mfg. Efficiency

Source: Jake Callahan | CNBC | July 21, 2023 | Link

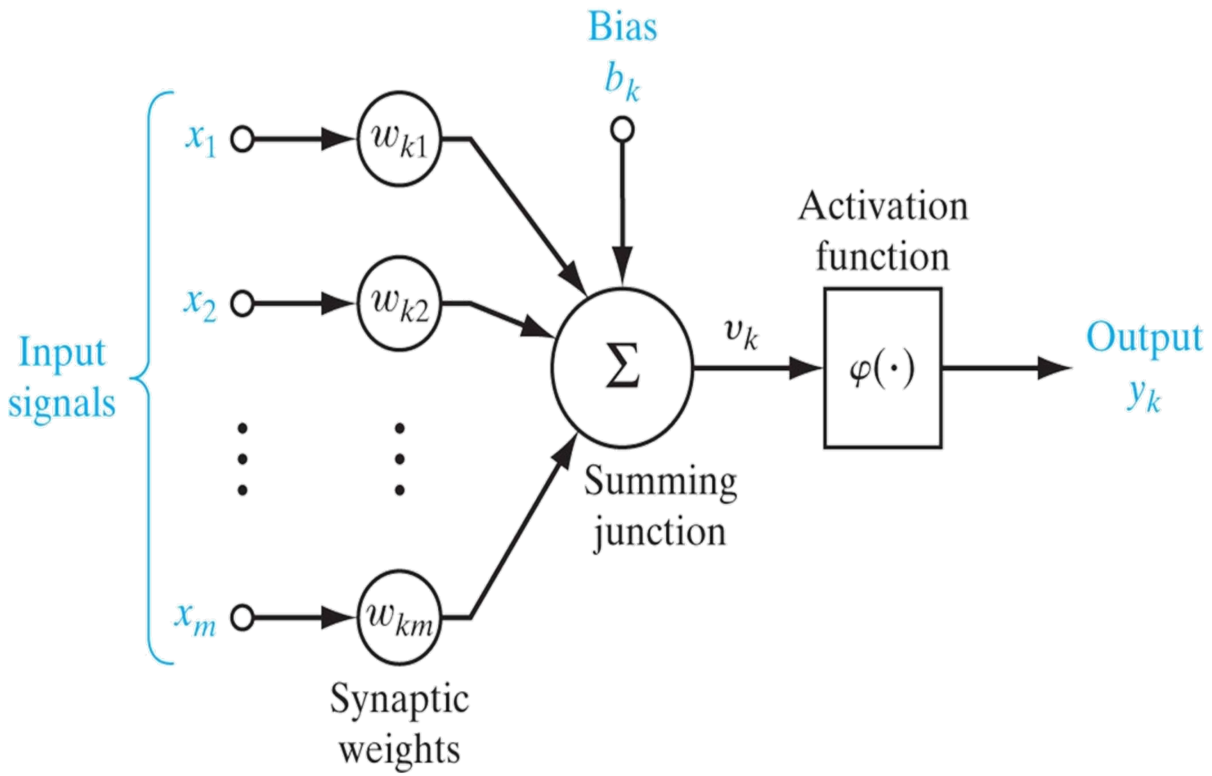
Definition of an ANN (Haykin, 1994)

- **A "massively" parallel distributed processor** that has a natural property for storing experiential knowledge and making it available for use.
- **Employs "massive" interconnection** of **simple computing cells** referred to as **"neurons"**
- Resembles the brain in two aspects:
  - ***Knowledge is acquired*** by the network through a ***learning process***.
  - Interneuron connection strengths known as ***synaptic weights are used to store the knowledge***.
- **Learning Process**: Procedure/algorithm used to modify the synaptic weights of network in an orderly fashion to attain a desired design objective

(Simple) Model of a Neuron

Three basic elements:

1. A set of ***synapses*** or connecting links, each of which is characterized by a *weight* or strength of its own.
   - Weight is +ve if associated synapse is excitatory; it is –ve if synapse is inhibitory.
2. An ***adder*** for summing input signals, weighted by respective weights.
3. An ***activation*** *function* for limiting the amplitude of output of a neuron and building non-linearity into the network.



Neuron Model

Source: Simon Haykin (2009)

(Simple) Model of a Neuron

Mathematical Terms:

1. **Linear Combiner Output**:

$$u_k = \sum_{j=1}^{m} w_{kj} x_j$$

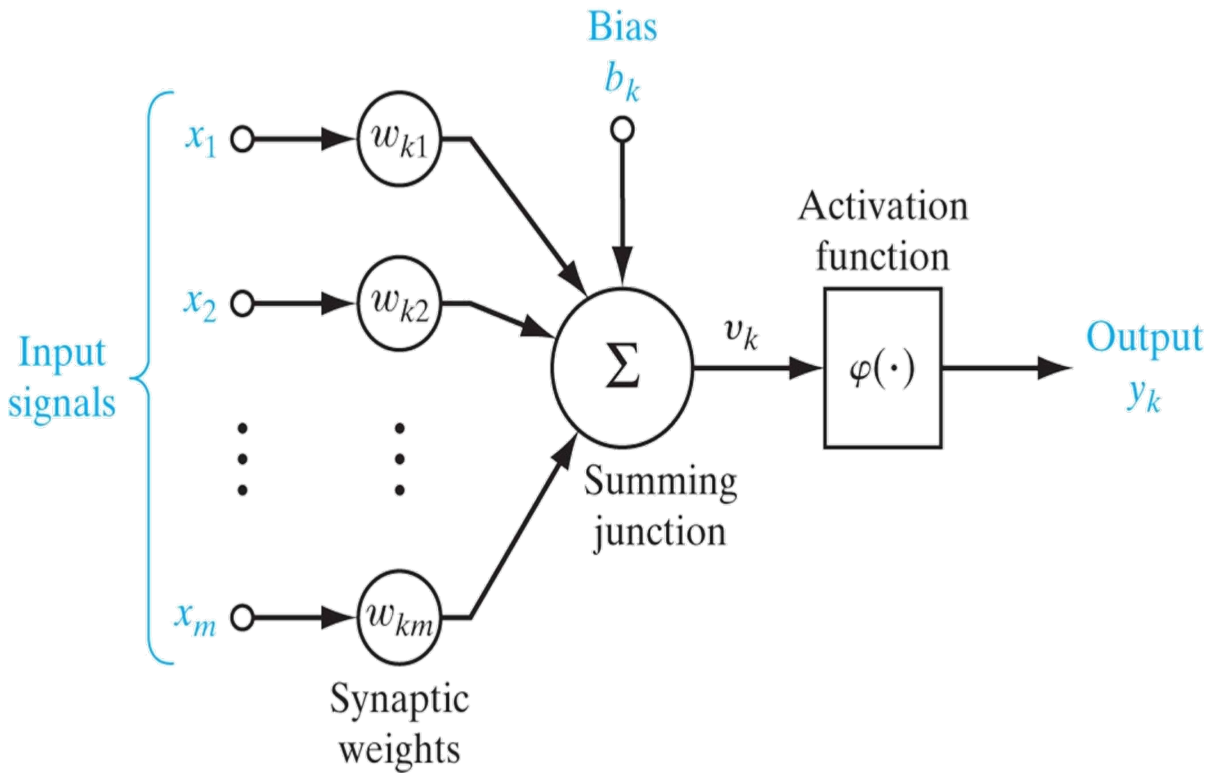
2. **Activation Potential**:

$$v_k = u_k + b_k \quad \text{or} \quad v_k = \sum_{j=0}^{m} w_{kj} x_j$$

where $x_0 = 1$ and $w_{k0} = b_k$

3. **Neuron Output**:

$$y_k = \varphi(v_k)$$



Neuron Model

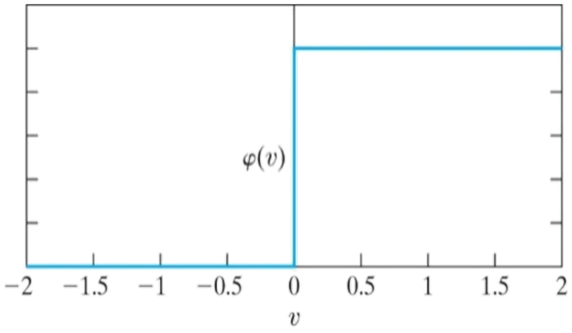
Bias allows the net activation potential to be nonzero even if all the inputs are zeroes.

Activation Functions

- **Defines neuron output in terms of activity level at its input**
- Some popular types: 1) Threshold (Hard-Limiter) Function, 2) Piecewise-Linear Function, 3) Sigmoid Function, 4) Radial Basis Function, 5) Rectilinear Unit, and 6) Softmax Function.

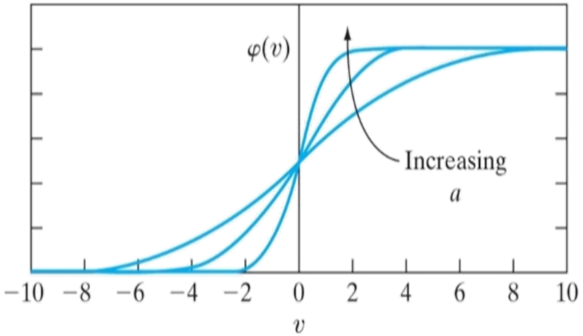
Threshold Functions:
An example:

$$\varphi(\vartheta) = \begin{cases} 1 & if \vartheta \geq 0 \\ 0 & if \vartheta < 0 \end{cases}$$



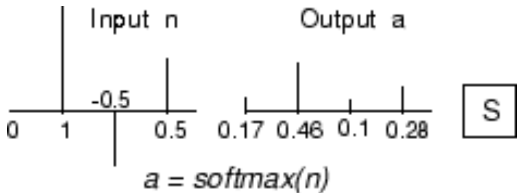
Sigmoid Functions:
Differentiable, strictly increasing, smooth, with asymptotic properties.
Example: Logistic function

$$\varphi(\vartheta) = \frac{b}{1 + e^{-a\vartheta}}$$



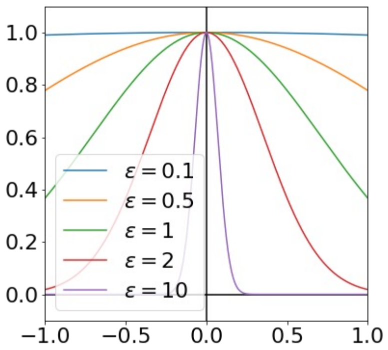
Softmax Function:
Generalization of logistic function that "squashes" a *K*-dimensional vector of arbitrary real values to a *K*-dimensional vector of values in the range (0,1) and add to 1:

$$P(a_j|\boldsymbol{n}) = \frac{e^{n_j}}{\sum_{k=1}^{K} e^{n_k}} \text{ for } j = 1, \ldots, K$$



Activation Functions

Radial Basis Function:
A function that depends only on distance from the origin or some point c, so that $\phi(x, c) = \phi(\|x - c\|)$.
Norm is usually Euclidean. Commonly used examples include (writing $r = \|x - x_i\|$):



- Gaussian:
  The first term—that which is used for normalisation of the Gaussian—is missing because every Gaussian has a weight in our sum, thus the normalisation is not necessary.

  $$\phi(r) = e^{-(\varepsilon r)^2}$$

- Multiquadric:

  $$\phi(r) = \sqrt{1 + (\varepsilon r)^2}$$

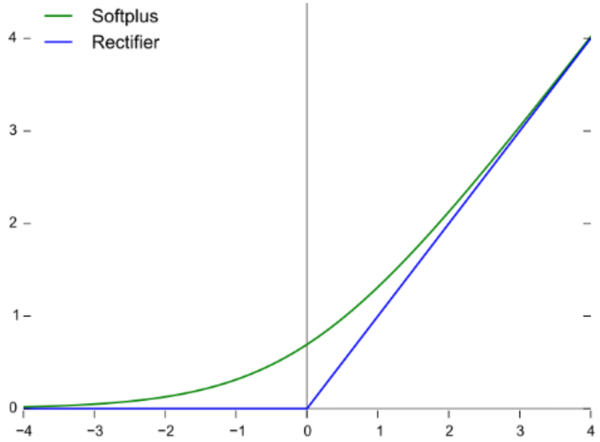
- Inverse quadratic:

  $$\phi(r) = \frac{1}{1 + (\varepsilon r)^2}$$

- Inverse multiquadric:

  $$\phi(r) = \frac{1}{\sqrt{1 + (\varepsilon r)^2}}$$

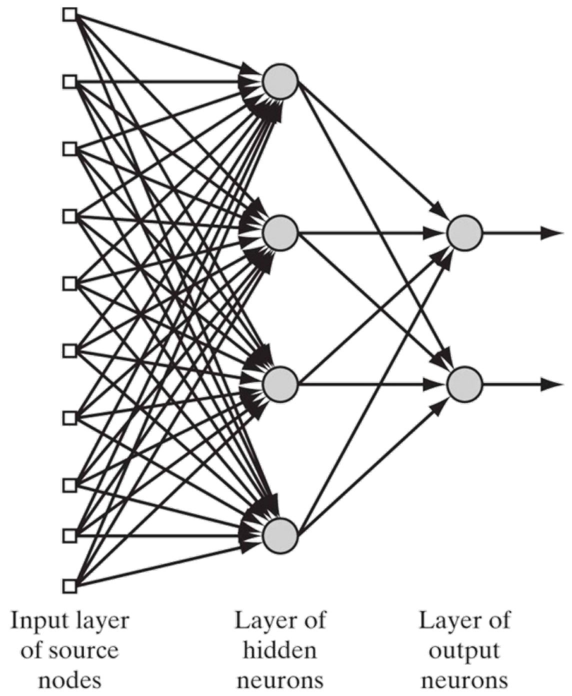
Rectilinear Unit (ReLU):
A function that retains the positive part of its argument. There are several variants.



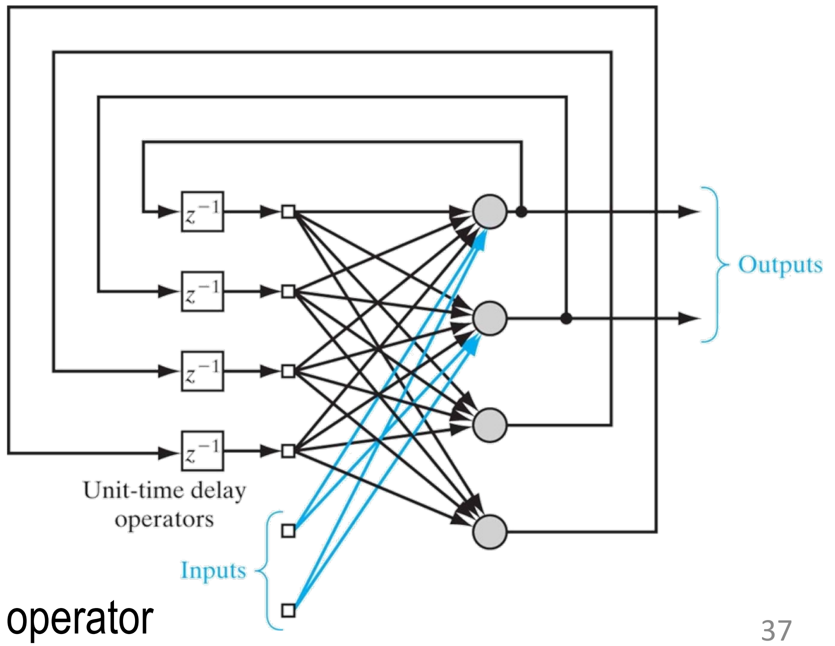
ANN Architecture

- **Def.: How neurons of a network are connected**
- Two classes of architectures:
- ***Feed-forward Networks***
  - Neurons are organized in layers
    - Input layer, multiple hidden layers, and output layer
  - Information strictly flows forward from input layer to output layer
- ***Recurrent Networks***
  - Has at least one feedback loop
  - Have strong dynamic properties

Applications: Function approximation, pattern recognition, time-series forecasting etc.



Applications: Modeling dynamic systems and optimization



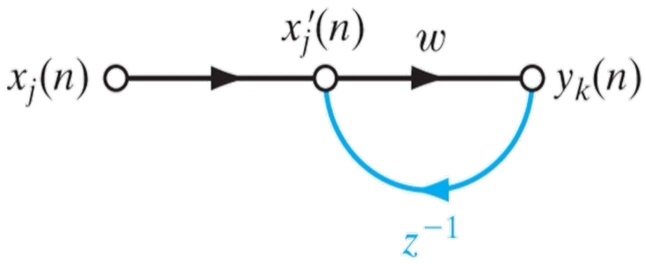
$z^{-1}x(n) = x(n-1)$

z^{-1} denotes a unit-delay operator

Source: Simon Haykin (2009)

Feedback

- **Exists when output of an element in system influences in part input applied to that same element**
- Plays a major role in the study of recurrent networks
- Consider the following single-loop feedback system:

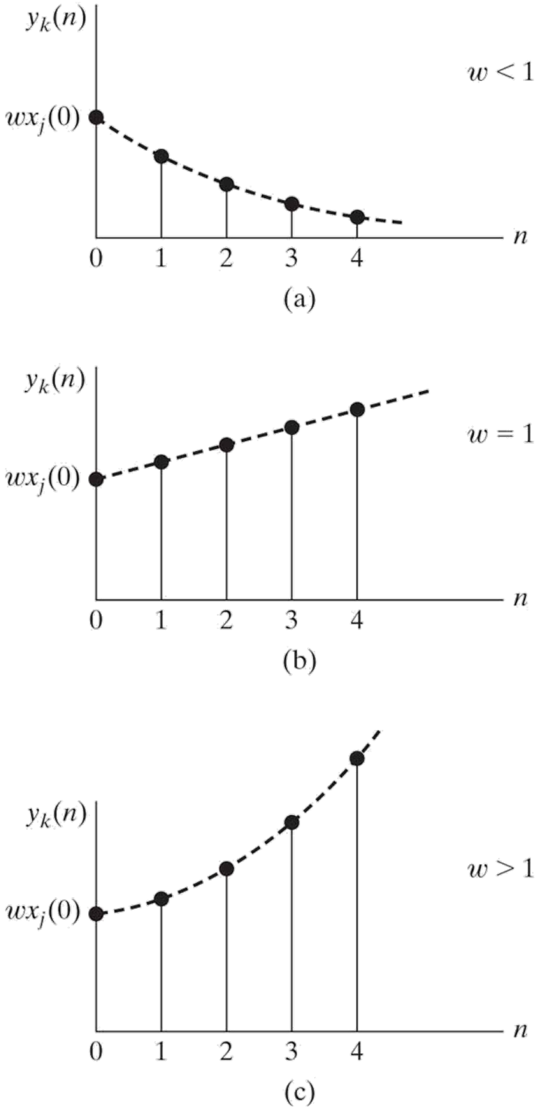


$$y_k(n) = w\left[x_j'(n)\right]$$
$$x_j'(n) = x_j(n) + z^{-1}y_k(n)$$

$$\Rightarrow \quad y_k(n) = \frac{w}{1-wz^{-1}}\left[x_j(n)\right] \quad \Rightarrow \quad y_k(n) = w\sum_{l=0}^{\infty} w^l z^{-l}\left[x_j(n)\right]$$

Binomial Expansion

Source: Simon Haykin (2009)



Time response for three different values of weight w. (a) Stable with infinite memory. (b) Linear divergence. (c) Exponential divergence.

Example: Fit Simple Regression Model

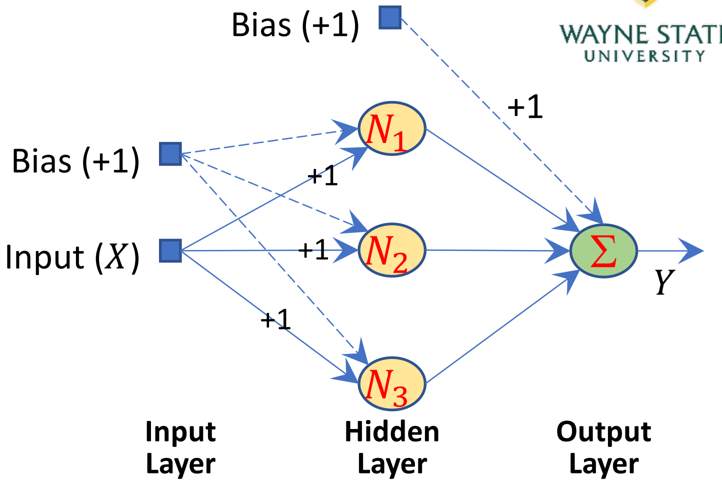
- **Network**: 1 Input & 1 Output
- Hidden Layer Neurons: 3
- Transfer Function:
  - Hidden: Logistic
  - Output: Identity

Logistic Transfer Function:

$$\varphi(\vartheta) = \frac{b}{1 + e^{-a\vartheta}}$$

Scale Parameter

Shape Parameter



Setting: #1

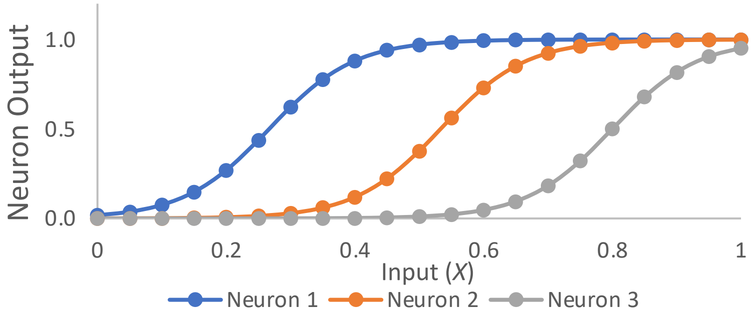
Hidden Neuron	N_1	N_2	N_3
Bias Weight	**4**	**8**	**12**
Shape Parameter	15	15	15
Scale Parameter	1	1	1
Output Weights	**-1**	**1**	**1**

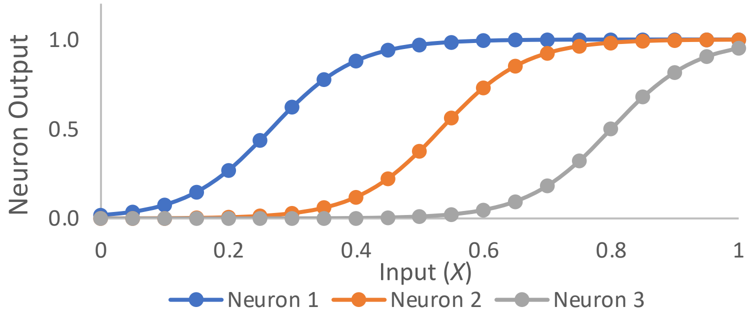
Setting: #2

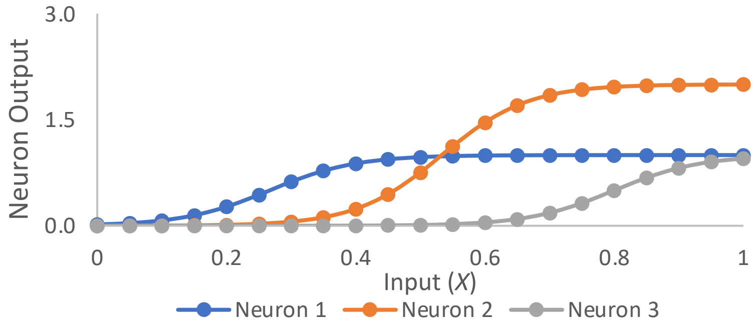
Hidden Neuron	N_1	N_2	N_3
Bias Weight	4	8	12
Shape Parameter	15	15	15
Scale Parameter	1	1	1
Output Weights	-1	1	**-1**

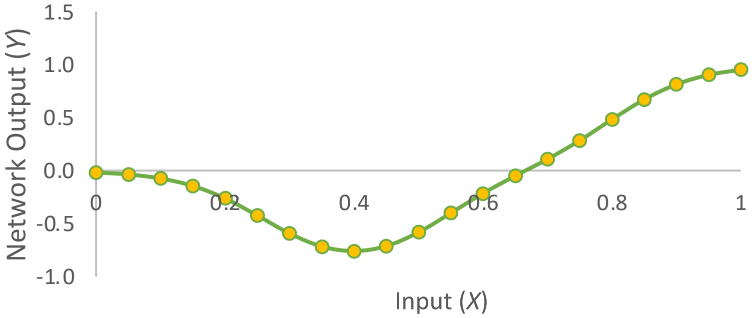
Setting: #3

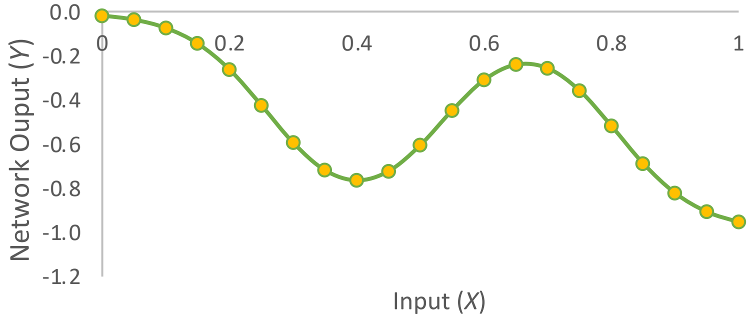
Hidden Neuron	N_1	N_2	N_3
Bias Weight	4	8	12
Shape Parameter	15	15	15
Scale Parameter	1	**2**	1
Output Weights	-1	1	1

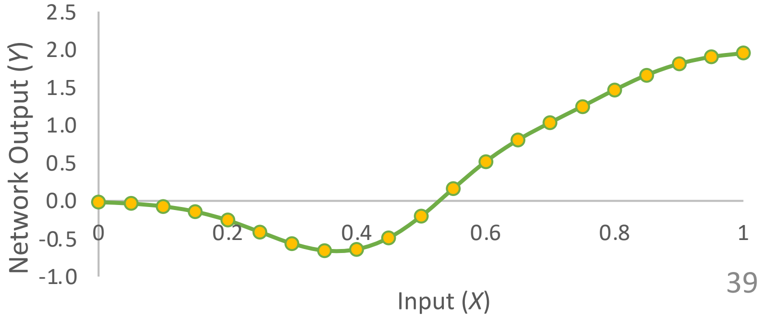












Knowledge Representation in ANNs

- Knowledge can be classified into two kinds of information:
  - Known facts (*prior* information)
  - Observations and measurement (inherently noisy)
- Complicated topic that can only be addressed effectively one application at a time
- **Network Design:** Generally, involves these steps:
  - Selection of appropriate architecture
  - Collection of example patterns representative of environment (both +ve and -ve examples)
  - Training network by means of a suitable algorithm
  - Testing network performance for generalization

Source: Simon Haykin (2009)

Knowledge Representation in ANNs …

- Four general rules (pattern classifier metaphor) (Anderson, 1988):
    1. Inputs from "similar" classes should usually produce similar representations inside network
    2. Items from "separate" classes should be given widely different representations in the network
    3. Important features should be allocated a larger number of neurons in the network for that feature's representation
    4. *Prior information* and *invariances* should be built into the design of a neural network and not “learn” them
        - Results in fewer network parameters and a faster/cheaper network
- Image Recognition Example:
    - Prior Information – Two-dimensional image, BW/Color, Critical Features, …
    - Invariance Requirements – Invariance to rotation, size, location, sharpness, brightness, …

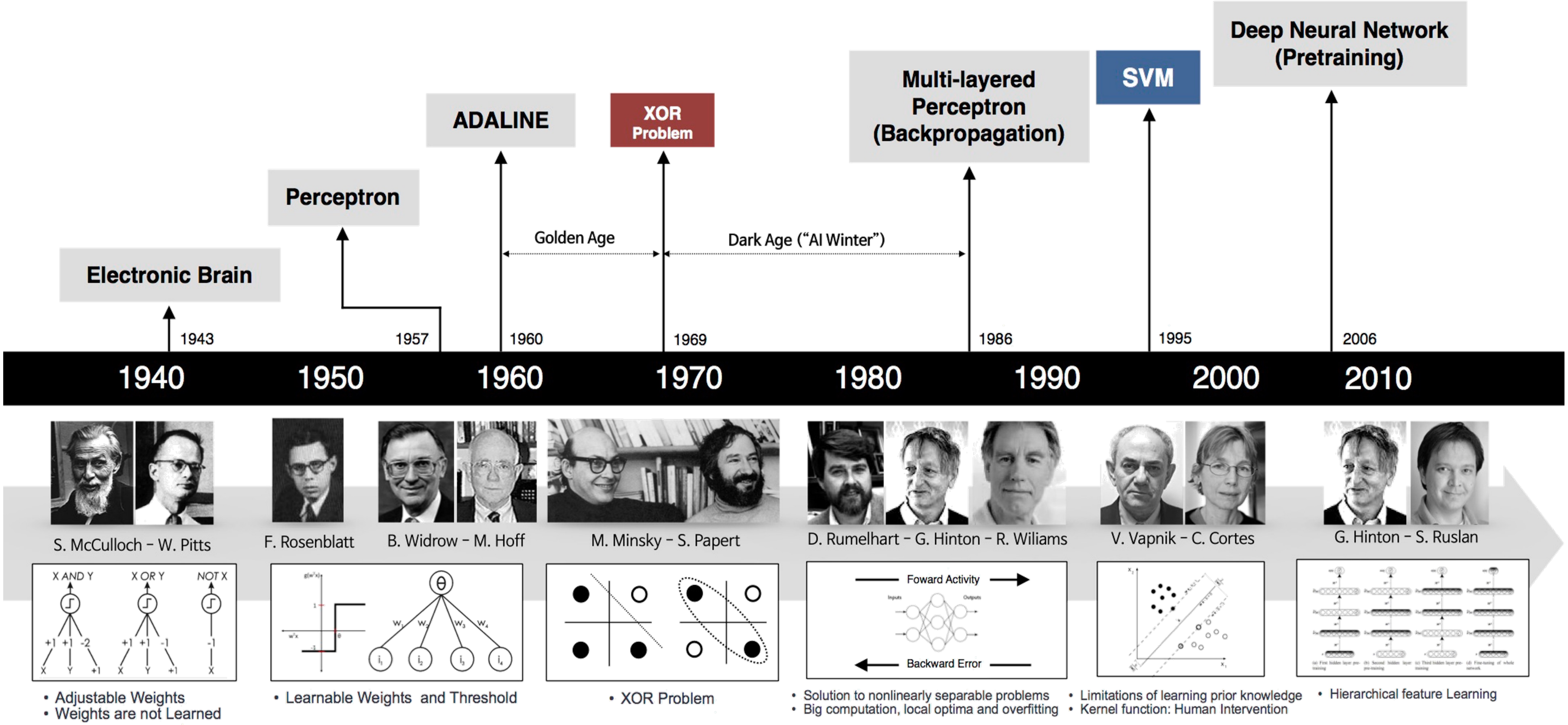
Source: Simon Haykin (2009)

Knowledge Representation in ANNs …

- In general, “prior” information can be built into ANNs in two ways:
  1. Through network architecture restrictions
  2. Through synaptic weight constraints
- In general, “invariances” can be built into the ANNs in three ways:
  1. Through structure (architecture restrictions and synaptic weight constraints)
  2. Through training
  3. Through invariant feature space

Source: Simon Haykin (2009)

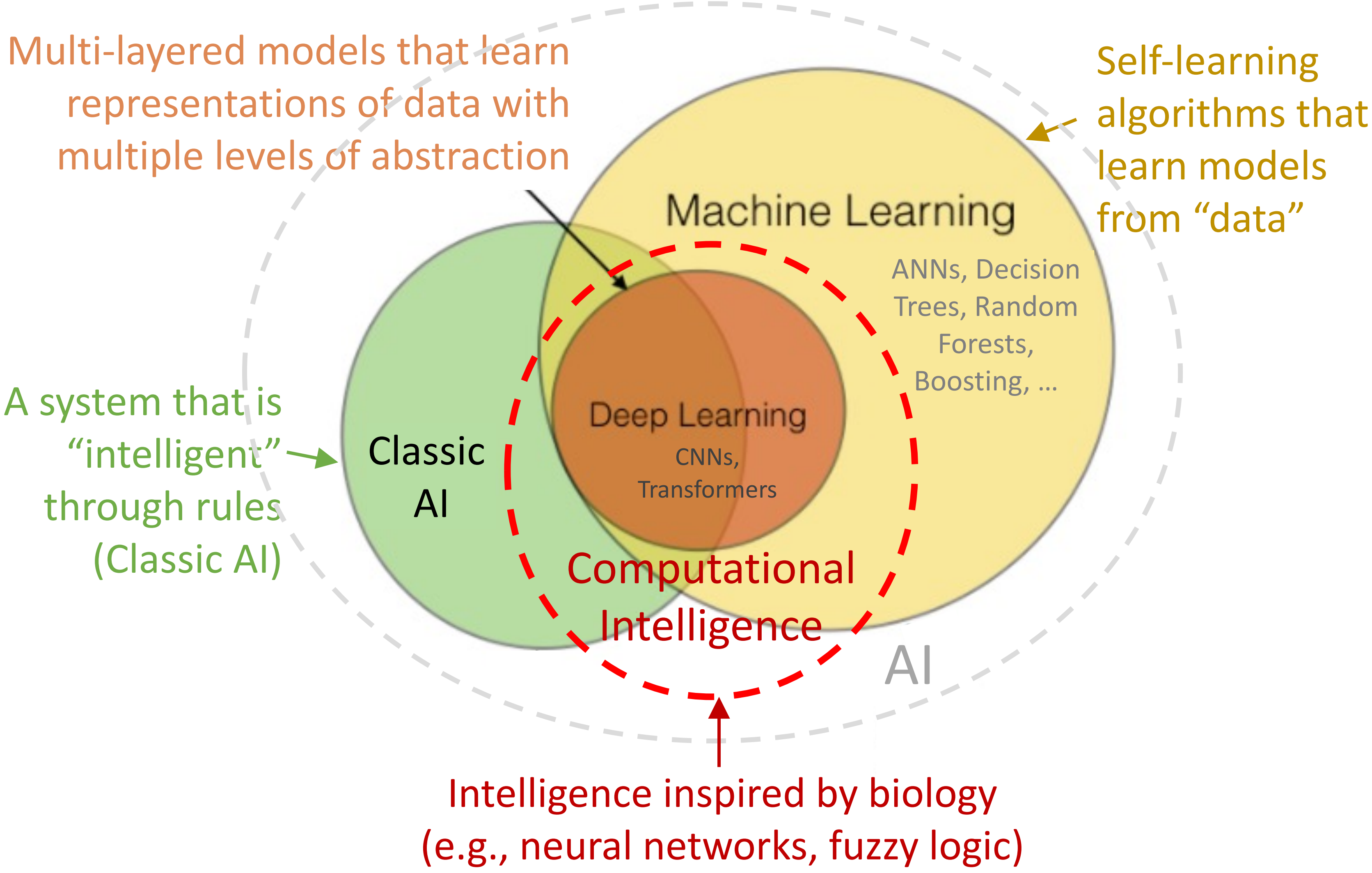
Neural Networks Timeline



SOURCE: Unknown

AI vs Machine Learning vs CI vs Deep Learning

- **Artificial Intelligence**:
  - The field of developing computers and robots that are ***capable of behaving in ways that both mimic and go beyond human capabilities***.
- **Machine Learning**:
  - Subcategory of AI uses algorithms to ***learn insights and recognize patterns from data***, applying that learning to make increasingly better decisions.



AI: Artificial intelligence
CI: Computational intelligence

Source: J. Chu, Understanding AI, Machine Learning, and Deep Learning | [LINK]()

Neural Network Hardware: *Examples*

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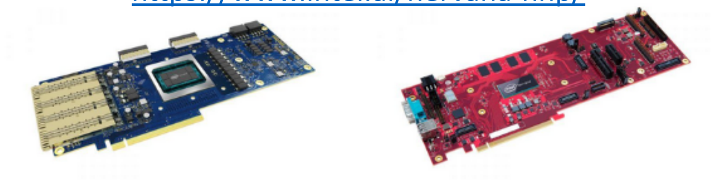
https://www.intel.ai/nervana-nnp/



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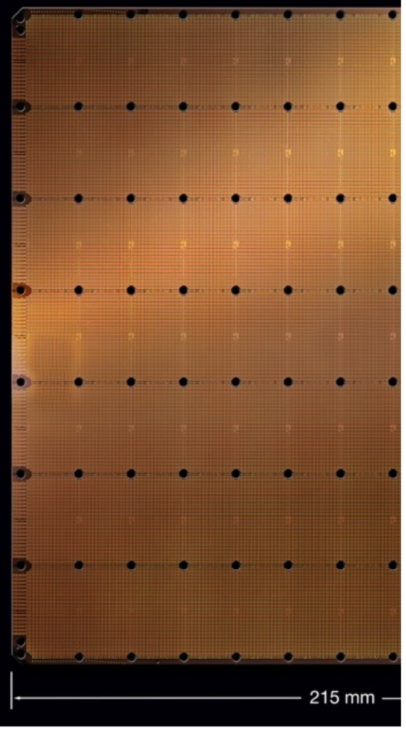

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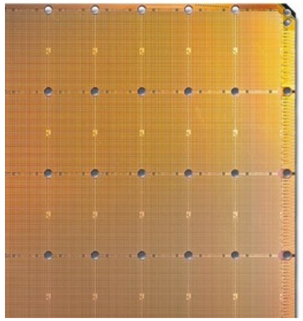

The WSE is the largest chip ever built

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Challenges with Deep Learning & AI

Jerome Pesenti, VP of AI @ Facebook

Source: Wired Interview with J Pesenti, Dec 2019 | LINK

- IBM Watson called out that **this is a commercial market** and that there are applications. It was remarkable. But **there was too much overhyping**. I don't think that served IBM very well.
- **Deep learning and current AI has a lot of limitations**. We are very far from human intelligence. It can propagate human biases, it's not easy to explain, it doesn't have common sense, it's more on the level of pattern matching than robust semantic understanding.
  - But we're making progress and the field is still progressing pretty fast.
- **AI "reproducibility crisis,"** or the difficulty of recreating groundbreaking research. It's something that Facebook AI is very passionate about. When people do things that are not reproducible, it creates a lot of challenges. It's a lot of lost investment.
- OpenAI noted that **compute power required for advanced AI is doubling every 3.5 months**.
  - When you scale deep learning, it tends to behave better and to be able to solve a broader task in a better way. But clearly the rate of progress is not sustainable. Nobody can afford that.
  - We really need to think in terms of optimization, in terms of cost benefit, and we really need to look at how we get most out of the compute we have. This is the world we are going into.

Good Article: Gary Marcus, An Epidemic of AI Misinformation, The Gradient, 30 Nov 2019 | Link

AI pioneer Geoffrey Hinton leaves Google:

Warns of technology's dangers!

Mr. Hinton is leaving Google so that he can freely share his concern that AI could cause the world serious harm.

JOE CASTALDO | PUBLISHED MAY 1, 2023 | [Globe & Mail]

- Geoffrey Hinton has quit his position at Google in order to speak freely about the dangers posed by AI, saying that part of him regrets his life's work.
  - He has served as a vice-president and engineering fellow.
- Chief among his concerns is that generative AI systems can overwhelm the internet with fake photos, videos and text, impairing people's ability to differentiate truth from fiction.
- **Mr. Hinton is worried about job losses and the potential for AI to outsmart humans in the longer term.**
- "The idea that this stuff could actually get smarter than people – a few people believed that. Most people thought it was way off. And I thought it was way off. … Obviously, I no longer think that." – Hinton (NY Times)

On-Demand Web Based Access to Grid Computing

Example:
Jupyter Notebooks for Python & R with GPUs

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